Agentic Statistical Analysis Assistant

CS7327 | Midterm writeup | Current implementation and limitations

What we built

We built a local English-language assistant that accepts a CSV and a natural-language research question, produces a structured analysis plan, executes statistical tools and explains the results with uncertainty and limitations. The implemented workflow is an actual LangGraph StateGraph with conditional routing, user clarification, bounded feedback and short-term session memory. It currently supports Welch independent-samples t-tests, Pearson correlation, chi-square tests of independence and simple ordinary least-squares (OLS) regression. It does not automatically search for alternative methods or claim general-purpose statistical analysis.

Agent roles and why they are separate

Planner (DeepSeek): maps the question to variable roles, goal, study design and any date filter; requests clarification when required information is missing. Interpreter (DeepSeek): turns the plan, computed results and diagnostic warnings into a cautious explanation. These are two distinct LLM roles using the same model. Shared graph state carries the Planner's plan through the statistical tools to the Interpreter; this is the agent handoff.

Python nodes profile the data, select a supported method using explicit rules, check assumptions and call SciPy. A deterministic critique checks placeholders, numerical formatting, method wording and selected unsafe claims; it is not an additional LLM agent. Language interpretation benefits from an LLM, while reproducible rules and numerical calculations belong in code.

One end-to-end walkthrough

Input: the built-in synthetic two-group dataset has 50 observations, 25 treatment and 25 control, with columns group and outcome. The question is "Compare mean outcome between treatment and control." The Planner identifies goal=difference, outcome=outcome and group=group. The user confirms independent observations and an observational study. These design facts cannot be reliably inferred from column values alone.

The rule node selects Welch because the question concerns a mean difference between two groups with a continuous outcome. Pearson would instead answer a question about linear association between two numerical variables. Welch does not require equal variances. Diagnostics examine complete finite values, group counts, nonconstant values and available Shapiro–Wilk, skewness and outlier checks. These checks do not prove independence or guarantee that inference assumptions hold.

SciPy estimates treatment minus control as 1.2532, with a 95% confidence interval of approximately [0.8176, 1.6888] and $p \approx 5.797 \times 10^{-7}$, using all 50 observations. The Interpreter describes evidence of an observed group difference and uncertainty, without claiming causation or practical importance. Numerical placeholders are filled from tool results. The critique can trigger one rewrite; persistent failure withholds the explanation. Unusable data stops with an issue report. Plan-format repair is also limited to one attempt, with an overall five-call model budget; there is no unbounded method-search loop.

Follow-up questions and memory

Within the same thread, “Explain the result” reuses the stored plan, statistics and warnings and requests a new explanation; it does not rerun the statistical test. In contrast, “Repeat for 2025” reuses the variable roles and confirmed design but applies a new date filter and recomputes the subset, diagnostics, sample size, statistical result and interpretation. For the built-in correlation example, 20 observations across 2024–2025 become 10 observations in 2025; Pearson r changes from approximately 0.9929 to 0.9755. Old numerical conclusions are not reused after filtering. Ambiguous phrases such as “last year” require an explicit period rather than a silent guess.

What “basic statistical checks” means here

All methods check relevant column types, usable observations and missing or invalid values. Welch checks each group’s size and distribution diagnostics. Pearson checks nonconstant numerical variables, sample size and distribution/outlier warnings; these do not establish linearity. Chi-square checks category counts and expected cell frequencies and stops for an unsupported sparse table. Simple OLS checks predictor variation, residual normality, heteroscedasticity and influential points, reporting warnings. Paired or clustered designs and multivariable regression are outside the current scope. Failed checks may stop analysis or generate explicit limitations; this version does not automatically switch to a nonparametric method.

A genuine breaking scenario

In a controlled failure test, we inserted “This means the outcome increases by eighty percent” into an otherwise correctly structured Pearson interpretation. The current code accepted it. This is statistically wrong: a correlation coefficient measures the strength and direction of linear association, not a percentage increase in an outcome. The sentence was an authored test mutation, not a claim generated by DeepSeek; the observed failure belongs to our checker.

The checker focuses on numerical placeholders, formats and selected phrases, so words such as “eighty percent” can escape its checks. Fixing this requires method-specific semantic tests and a review rubric for effect meaning, overstatement and limitations. Independent teammate review is proposed: each reviewer sees the question, tool output and explanation, marks acceptable/misleading/uncertain and records the problematic sentence and reason. Disagreements are resolved after independent labeling. An AI reviewer could assist, but cannot establish reliability without comparison against those labels. These additions are planned, not completed.

Evaluation and what the evidence supports

The benchmark contains 31 predefined cases: 4 ordinary, 10 borderline, 4 clarification, 11 stop and 2 follow-up cases. The first live batch met strict mechanical expectations in 28/31 cases; the three mismatches concerned safe-stop or clarification behavior. The 17 computed results passed numerical-reference comparisons. Correct arithmetic does not prove the selected assumptions or interpretation are valid.

A separate controlled challenge set contains 4 valid interpretations and 12 deliberately planted errors. The code accepted all 4 valid templates and caught 9/12 planted errors, missing 3 semantic mistakes. This measures detector coverage, not real-model accuracy. No human semantic accuracy rate has been claimed. The strongest current evidence is reproducible execution and numerical agreement; the weakest area is completeness of semantic interpretation review.